Sentinel SARI surveillance in Belgium in times of COVID-19 pandemic

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Background



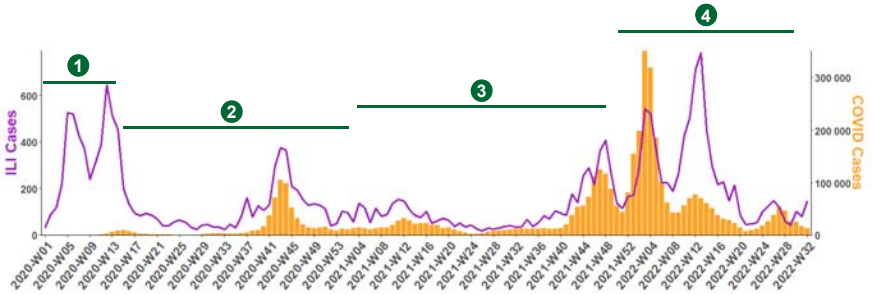
BELSARI-net
- 6 hospitals
- operating since 2011

SARI case definition* (pre-COVID)
- fever (or history)
- cough or dyspnea
- at least overnight hospitalisation
- symptoms before admission and within 10 days before sampling

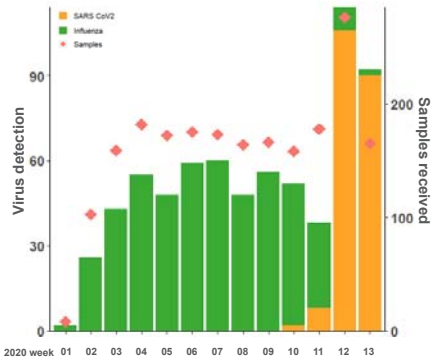
** based on WHO case definition*

COVID-19 pandemic phases and ILI-incidence rate (per week, Jan-2020 to Jul-2022)

- COVID cases: number of cases recorded and reported to ECDC/WHO-EURO
- ILI-incidence rate (per 100000 inhabitants) as calculated based on GP sentinel network reports



1 Early phase



- SARS-CoV-2 detection PCR implemented early Feb 2020 *(based on Corman, Eurosurv, 2020)*
- Normal influenza season from week 02 till 11
- Detection of 1st SARS-CoV-2 positive on week 10 (sampling date), same week as 1st local case reported by National COVID-19 monitoring platform
- Follow up of SARS-CoV-2 initial surge before lockdown

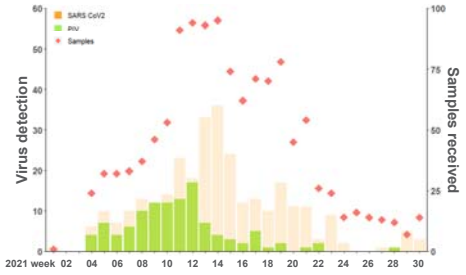
2 2020: First waves

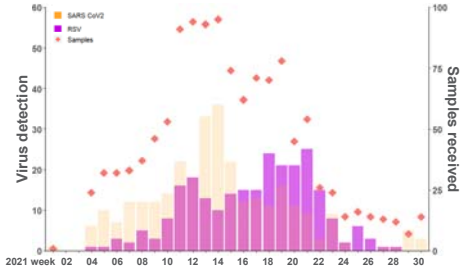
- Lockdowns and strong NPI* measures
- SARI surveillance interrupted (because of workload in hospitals)

** Non Pharmaceutical Interventions*

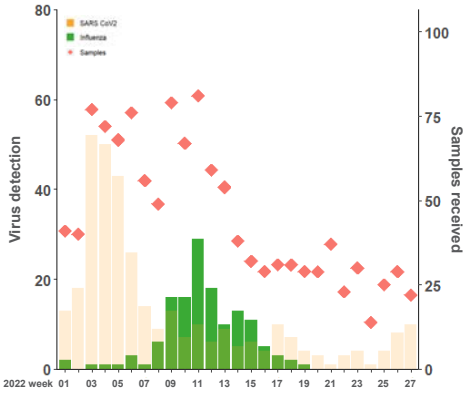
3 2021: PIV and RSV epidemics

- SARI surveillance able to resume, but in degraded mode *(not exhaustive recruitment)* and broader definition (symptoms), year round
- Follow up of SARS-CoV-2: Alpha wave (early 2021) followed by Delta wave (from week 35)
- Detection of various respiratory viruses other than SARS-CoV-2
- Parainfluenza (PIV-3) and Respiratory Syncytial virus (RSV-A) epidemics detected, but at unusual period for RSV
- No influenza epidemic





4 2022: Influenza is back!



- Recruitment improved as burden in hospital decreased
- Follow up of SARS-CoV-2: Omicron wave
- Influenza virus epidemic, but slightly delayed compared to pre-COVID seasons

CONCLUSION

- Good follow up of SARS-CoV-2 waves
- Year-round: important to follow new patterns of circulation for other respiratory viruses
- Collected data (patients' risk factors and complication during hospitalization): useful to evaluate relative burden of disease of the different viruses

ACKNOWLEDGEMENTS
The authors thank Ilham Fdillate, Reinout Van Eycken, Assia Hamouda and Mona Abady for technical assistance, and Sarah Houben for data analysis

FINANCING
Belgian Federal and Regional Health Authorities, ECDC (latest project: VEBIS)

THE PRESENTING AUTHOR HAS NOTHING TO DISCLOSE

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